

Barath Suresh

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EDUCATION

Arizona State University, Ira A. Fulton Schools of Engineering – Tempe, AZ

Master of Science in Computer Science

August 2025 – May 2027

GPA: 3.67/4.00

SASTRA Deemed University – Thanjavur, Tamil Nadu, India

Bachelor of Technology in Computer Science and Engineering

GPA: 3.176/4.00

SKILLS

Languages: C/C++, Java, Python, Dart, JavaScript, TypeScript, HTML, CSS

Backend & Microservices: Spring Boot, Spring Cloud Gateway, Spring WebFlux, Spring MVC, Node.js, Express.js, REST APIs, microservices, distributed systems

Messaging & Streaming: RabbitMQ, MQTT, WebSocket, FFmpeg, HLS video streaming

Auth & Security: JWT, OAuth 2.0, scope-based authorization, zero-trust access control, RBAC

AI & Analytics: Spring AI, Gemma 3 (LLM), log anomaly detection

DevOps & Observability: Docker, Prometheus, Grafana, Loki, Zipkin, CI/CD, logging, metrics, distributed tracing

Databases & Storage: MongoDB, MinIO (object storage), Firebase

Tools & Collaboration: VS Code, IntelliJ IDEA, Android Studio, PyCharm, Postman, Git, Agile Methodology

PROFESSIONAL EXPERIENCE

TATA ELXSI LTD

December 2024 – July 2025

Software Development Engineer

Chennai, India

- Built Java Spring APIs backed by MongoDB to process Excel-based employee datasets, reducing parsing time by 20% and streamlining workforce data ingestion
- Containerized services with Docker and integrated Prometheus & Grafana, improving CI/CD pipeline reliability and enabling real-time observability across six containerized services (Backend, Frontend, Prometheus, Grafana, Ollama with Gemma 3, and Loki)
- Automated log anomaly detection by containerizing a Spring AI + Gemma 3 service that summarizes and flags abnormal logs, reducing manual log triage and improving incident detection speed
- Led weekly code reviews in a 5–8 member Agile team, reviewing pull requests before merge and tightening quality gates, which reduced production bugs and improved overall code quality

TATA ELXSI LTD

January 2024 – June 2024

Software Development Engineer Intern

Bengaluru, India

- Built and containerized backend applications with Node.js, Express.js, and MongoDB, reducing deployment time by half and improving scalability
- Developed and dockerized a product catalog management microservice as a demo project, highlighting containerization and deployment best practices and serving as a reference for building scalable services
- Integrated real-time communication using MQTT and WebSocket, optimizing message handling and connection parameters to reduce dropped messages and improve synchronization reliability for real-time updates

PROJECTS

Prism – Distributed Video Streaming Platform, Independent Project

December 2025 – January 2026

- Architected an asynchronous, RabbitMQ-based microservices platform with eight Spring Boot services, reducing video upload latency by 70% by offloading FFmpeg transcoding to background workers that generate four HLS quality variants
- Built a reactive smart proxy with Spring WebFlux and AWS SDK v2, enabling 5× more concurrent streams through non-blocking I/O while enforcing zero-trust access to private MinIO storage
- Designed end-to-end observability across eight microservices with distributed tracing (Zipkin), metrics (Prometheus), and centralized logs (Loki), enabling faster root-cause analysis and reducing debugging time by 60%
- Implemented a multi-tenant identity system using JWT authentication and scope-based authorization, enforcing access control through custom Spring Cloud Gateway filters across all APIs

HexScript – Secure Note-Taking App, Independent Project

September 2023

- Built a Flutter-based secure note-taking app with offline support and AES-encrypted local storage, enabling seamless access to notes in low-connectivity environments while safeguarding sensitive content from unauthorized access
- Implemented fingerprint-based biometric authentication for secure note access
- Designed a Material You-inspired UI with dark mode, reminders, and theme-aware customization to enhance usability and personalization